

Milestone 4: Formula 1 Sprint and DNF Prediction

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1 Background

Formula One, or F1 for short, is the highest class of auto racing sanctioned by the Fédération Internationale de l'Automobile (FIA). Teams compete in constructing cars and racing at circuits that take place all across the world. The competition began in 1950 and continues on to this day. Throughout the years, F1 has gone through many changes. The latest of these changes has been the addition of the sprint race format in 2021, which is a shorter standalone race that occurs on the Friday or Saturday before the main Sunday race.

Two aspects of F1 racing that have received little attention for predictive learning are sprint results and DNF's. In this report we see how well methods that predict results in Grand Prix races can be applied to sprint results, and look into how well DNFs can be predicted, if at all.

Sprints may present a different predictive challenge because sprint races are roughly one-third the distance of a standard Grand Prix and lack mandatory pit stops; they rely heavily on raw pace and grid positioning. Similarly, DNFs introduce massive variance into the sport, as unpredictable crashes or mechanical failures can instantly ruin a driver's race outcome. By focusing on these two underexplored areas, this analysis aims to uncover new insights into F1 race predictability.

2 Dataset

The analysis in this report is based on the Kaggle Dataset 'Formula 1 World Championship (1950–2024)' compiled by Vopani.¹ The dataset contains various statistics from F1 races from the above years, and is organized into 14 csv files. The csv files that will be explored in this report are listed in the table below.

Data Table	Description
circuits.csv	Information about individual race tracks
constructors.csv	Information on different car manufacturers
drivers.csv	General driver information
qualifying.csv	Qualifying times before each race
races.csv	Information on individual races
results.csv	Results for all Grand Prix races
sprint_results.csv	Results for all Sprint races

The dataset contains data for over 1000 grand prix races, 800 drivers, and 200 constructors. However, since sprint races have started quite recently, it only includes data for sprints starting in the 2021 season. As a result, the sprint focused portion of this analysis filters the dataset down to the 2021 through 2024 seasons, which contains 90 total race weekends. Of those 90 weekends, 18 included the sprint format, providing 360 individual sprint driver-race entries for evaluation.

3 Previous work

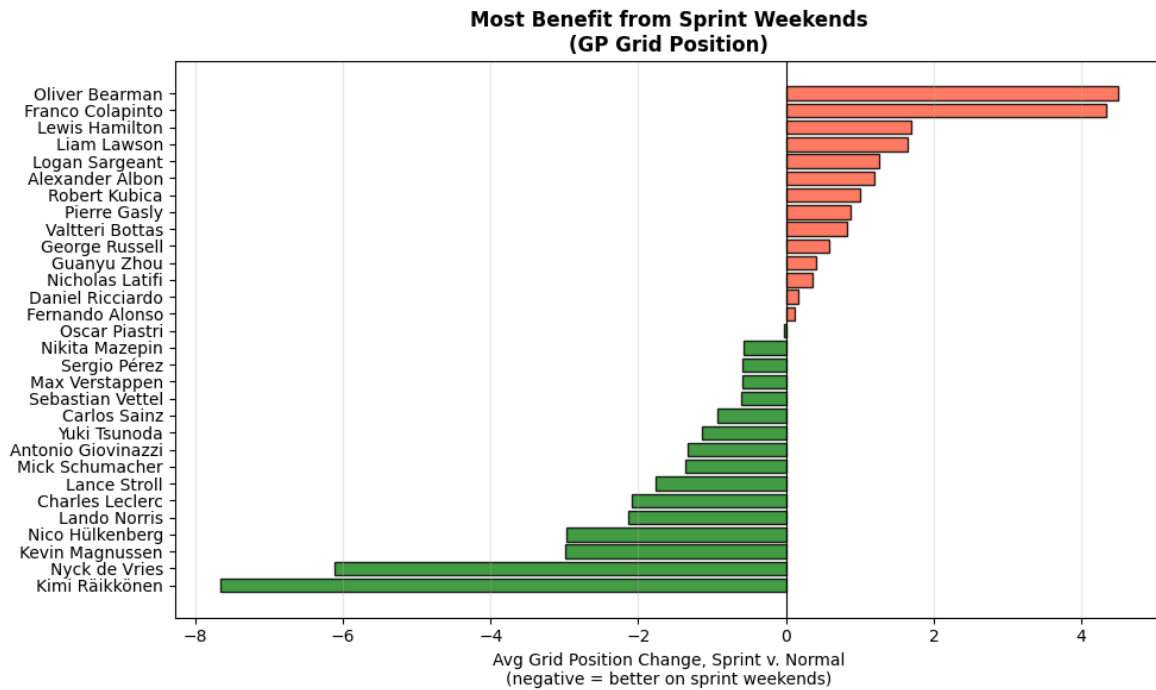
Our most referenced analysis was 'F1 Race & Qualifying Performance' by Jalel Gmiza^{1,2}. Previous analyses, such as this one, are focused on predicting Grand Prix winners and often extend that to predicting season winners. Sprints are nearly unmentioned, and DNFs are treated as anomalies, rather than potential targets. This is where our work branches off. We used some of the ensemble modeling techniques that are used in the reference analysis to inspire our own models, but adjusted them to either use sprint results or to contrast their results with sprint results.

4 Exploratory Data Analysis

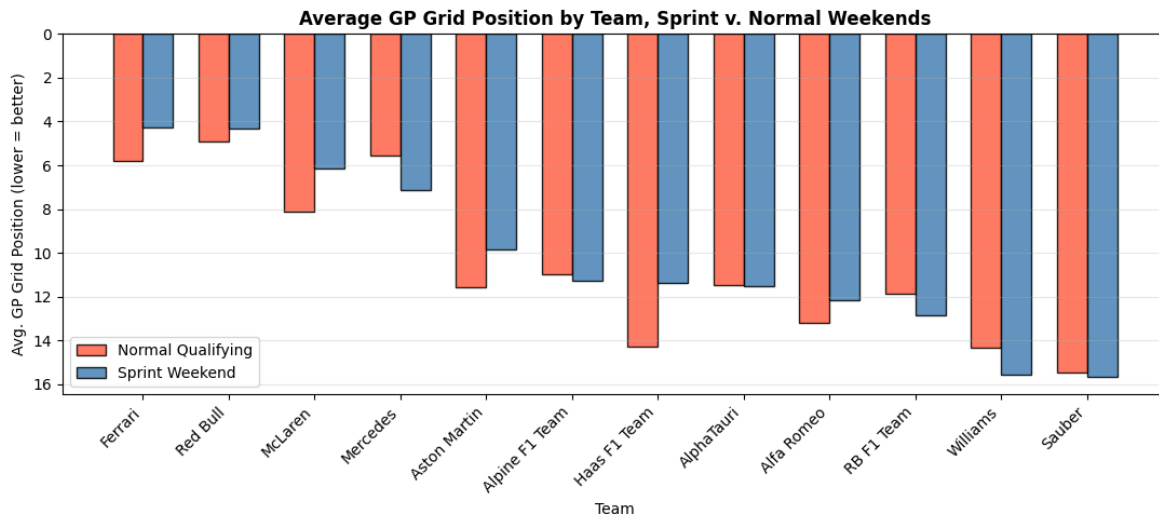
To identify optimal features for modeling, a comprehensive correlation analysis was performed. This study is structured into three primary analytical objectives: 1) Identifying factors that dictate sprint race outcomes, 2) Evaluating how sprint results influence subsequent Grand Prix performance within the same weekend, and 3) Determining variables that impact DNF occurrences. For clarity, the objectives will be labeled as Grand Prix Predictions, Sprint Predictions, and DNF Predictions, respectively.

Grand Prix Predictions

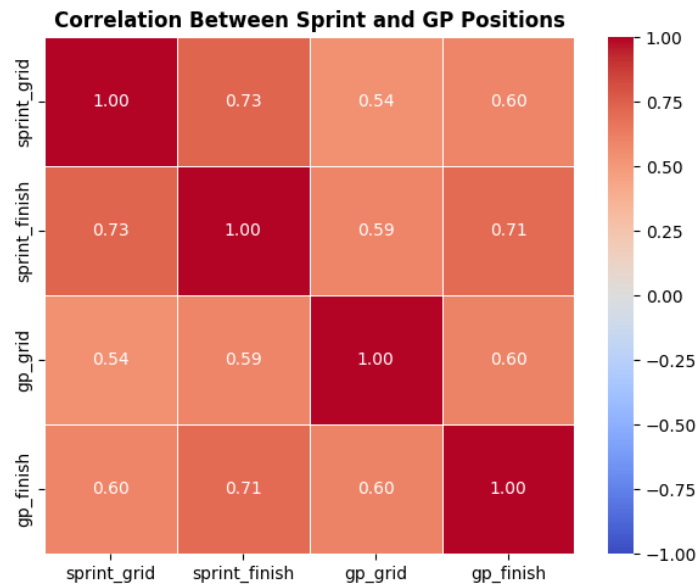
To properly evaluate the impact of the sprint, the dataset was filtered to include only race weekends from the 2021 to 2024 seasons. This filtering yielded 90 total race weekends, consisting of 18 sprint weekends and 72 normal qualifying weekends. With this subset isolated, an analysis was performed to determine whether Grand Prix grids dictated by sprint races differ from those set by standard qualifying sessions. By grouping the average GP grid positions by driver and format, a delta metric was calculated to reveal which drivers benefitted most from the sprint format. A more negative delta value indicated better GP grid positioning compared to a more positive delta value, indicating worse GP grid positioning. From the data, we saw that drivers such as Kimi Raikkonen (-7.67 delta) and Nyck de Vries (-6.11 delta) gained substantial grid advantages during sprint weekends. On the other hand, drivers such as Oliver Bearman (+4.50 delta) and Franco Colapinto (+4.33 delta) performed noticeably worse under this format.



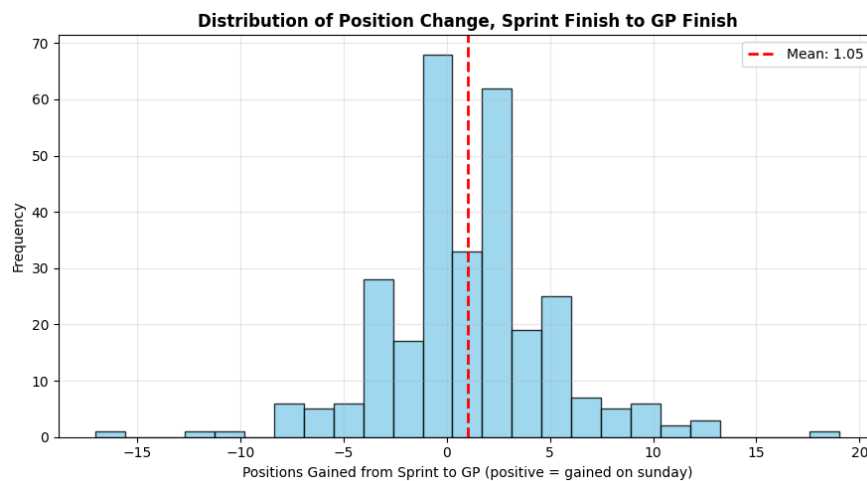
This variance was not just limited to individual drivers. Grouping the data by constructors demonstrated that the sprint format uniquely impacts different teams' setups and strategies as well.



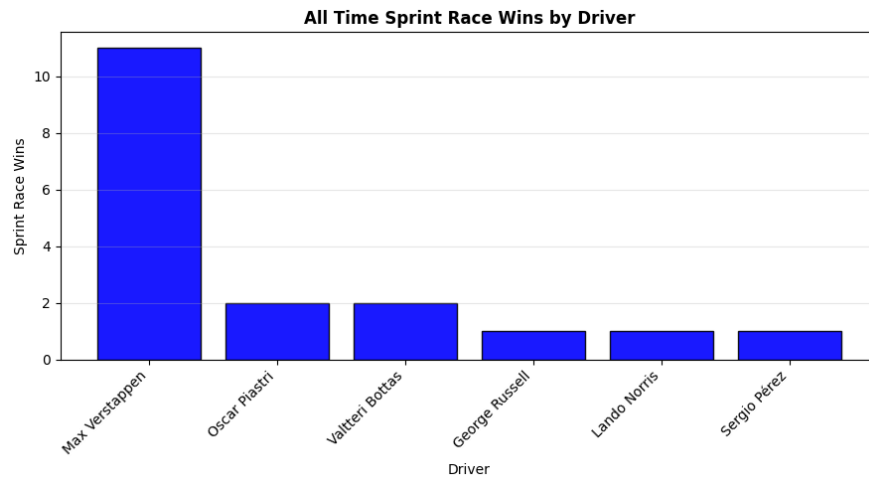
Furthermore, a correlation analysis was performed to examine the relationships between sprint grid position, sprint finish, GP grid position, and GP finish. The resulting matrix revealed that sprint grid positions heavily dictate sprint finishes, which subsequently share a strong correlation with the final Grand Prix outcomes.



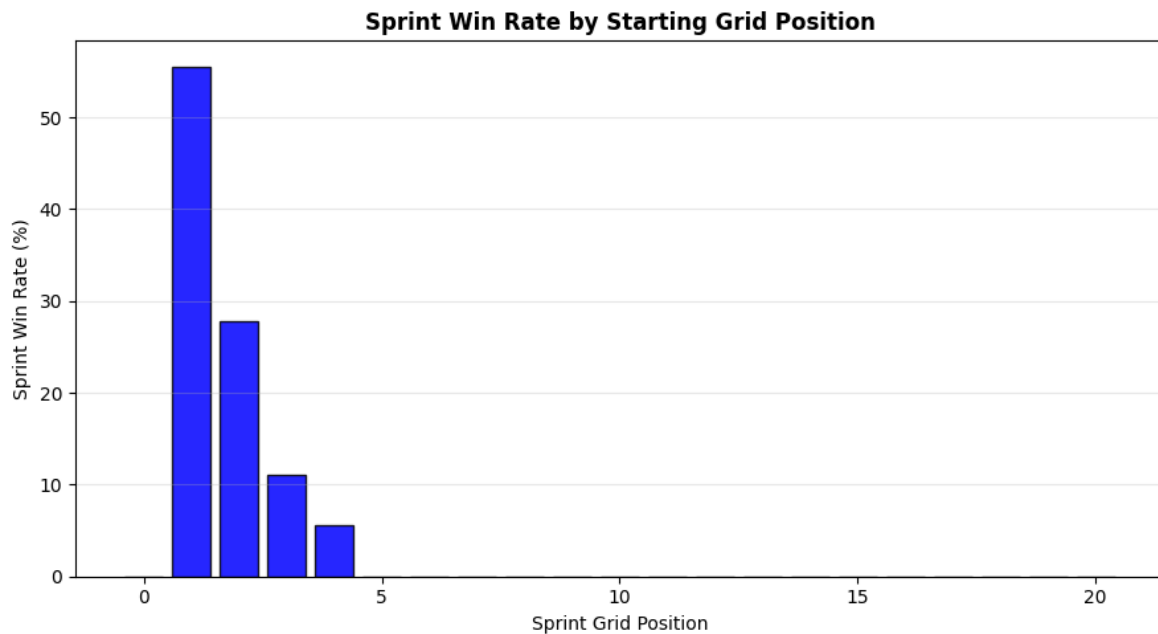
Examining the distribution of positions gained or lost between the sprint finish and the GP finish showed a concentrated distribution of a mean delta of 1.05. This slight positive shift indicates that, on average, a driver tends to gain about one position from their sprint finish to their final Sunday performance, though the tight clustering still reinforces that sprint outcomes are highly indicative of the overall Grand Prix results.



Moreover, we saw through analyzing sprint victories that there was a heavy concentration of wins among top drivers.



Plotting sprint win rates by starting grid position demonstrated that starting in position 1 is an overwhelming predictor of winning the sprint, with the probability of victory dropping sharply for any driver starting in position 2 or lower.



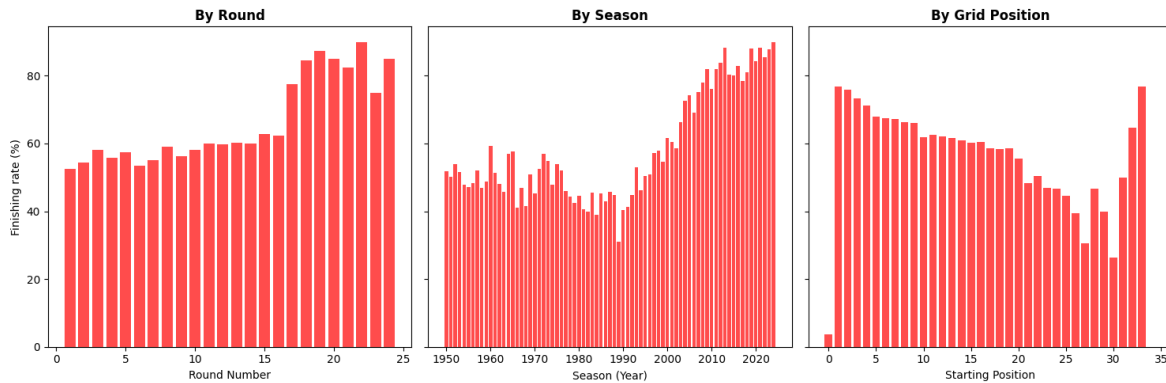
DNF Predictions

The EDA for DNFs is similar to the “F1 Race & Qualifying Performance”, but is not constrained by the qualifying times data, which only go as far back as 1994. Removing this constraint allowed for the entire history of F1 recess to be used in the analysis and modeling. This is important because DNF rates have varied widely based on the year.

	Total Records	Finished	DNF	Finishing rate
Total GP finishes	26,759	15,806	10,953	59.06%
GPs with Qualifying times	10,494	8,050	2,444	79.71%
Sprints	360	345	15	95.85%

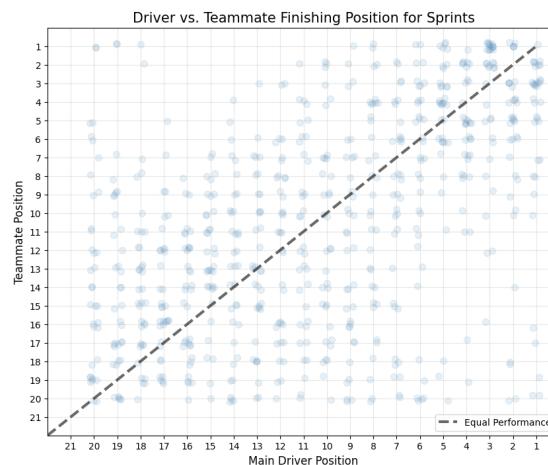
To assess how much of a difference using the varying DNF rates had on finishing positions, the starting and finishing positions in GPs were inspected first. In the full dataset, the mean positions changed was +3.0, in the limited dataset it was only +0.2. Because the number of drivers who finish is sometimes lower than the number that start, the drivers who do finish will have a better finishing position on average. Higher DNF rates exaggerate this effect.

Season (year), round, and starting grid position stood out as strong predictors of DNF rates. As F1 cars became more reliable, after a low in 1990, it also adopted longer seasons and smaller grids. This creates a skew in the data: results from higher round numbers reflect modern reliability standards, while results from the back of large, historical grids (e.g., 25th+) reflect the higher failure rates of the past. The plots below show these relationships, along with their very clear jumps and inflection points.



Sprint Predictions

The third section of the analysis related to data that would help predict the winners of the sprints. Some of the likely candidates to create features were car constructors, drivers, grid positions, and the track that the race is on. Each of these were tested to determine if they would be good features to include in the model. For drivers and constructors, to determine if the driver is more impactful or the constructor, the drivers placement compared to their teammates placement (another driver with the same constructor) was measured. The results are shown in the scatter plot below.



Although a little scattered, there is a clear correlation between the constructor of the car and the performance during the race. The calculated correlation coefficient was 0.51, meaning that the constructor of the car has some effect on the outcome of the race, and could be used as a metric to engineer features for.

5 Feature Engineering and Selection

Several features were engineered that were useful to prediction, EDA, or both. Here is a list of features that were used for DNF analysis and their purpose:

- `Completed_race` is a simple feature that gives a binary indication that individual racers finish the race based. If a racer's finishing position was NA in the results data then that racer was inferred to have not finished. This was used as a target metric.
- `Grid_size` to account for the number of drivers in the race. While most racers have 20 racers, some have more or less. Including this feature in the models reduced the number of impossible predictions, such as finishing 21st in a 20-person race.
- `prev_incomplete_count` as a cumulative total of races that team has not finished in that season. It is a reasonable expectation that if a team has already had mechanical failures in a season that they are prone to having them again in that season.
- `Positions_changed` was used in EDA, but not modeling. It indicates how many positions a racer gained or lost from their starting position in a given race. It could not be used for prediction, since it implies the correct prediction, but used to get an idea of the variance in the finishing positions.

To predict final positions for Sprint races, a few features based on cumulative driver and constructor performance were added:

- **Historical performance** - The average performance for a driver, gives a baseline on how the driver places on average.
- **Recent Performance** - A rolling window of the average of the past 5 races for the driver. This accounts for the current form of the driver and constructor, accounting for mechanical upgrades, and driver slump or new rises.
- **Circuit specific performance**: Drivers may find advantages driving through certain layouts, and this feature keeps track of the driver's position.
- **Cumulative position differential**: Gives the drivers volatility, and his chance of placing better or worse than his starting grid position.
- **Cumulative podium finishes**: The number of times a driver had already finished in the top three that season.

These are the features that were engineered to compare sprint and GP performance:

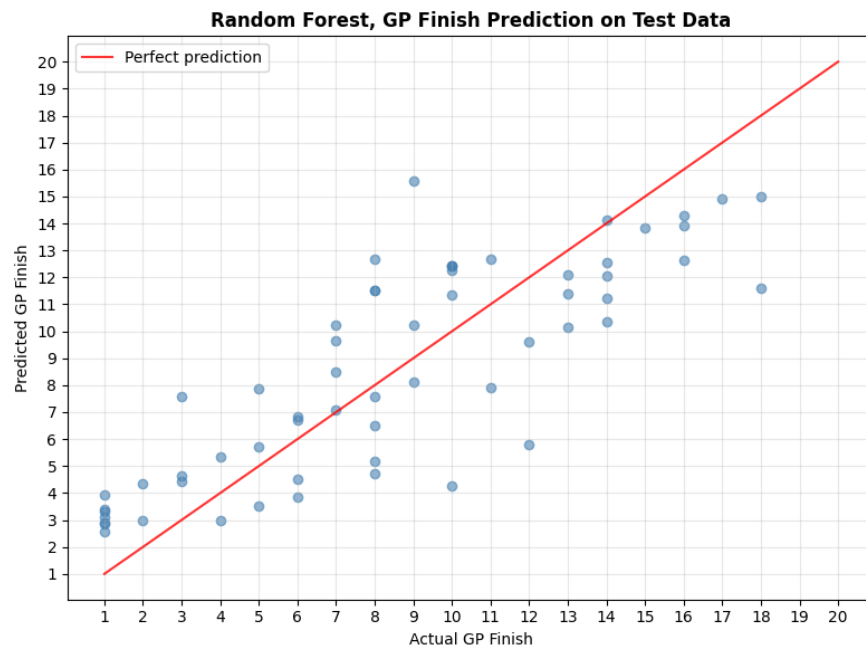
- `sprint_to_gp_delta` measures the numerical difference between a driver's sprint finishing position and their final GP finishing position on the following Sunday.
- `sprint_qualifying` and `normal_qualifying` were extracted to represent a driver's or team's average starting GP grid position under each respective qualifying format.

- `delta` captures the exact numerical difference between the aforementioned `sprint_qualifying` and `normal_qualifying` averages. This was used in EDA to highlight which specific drivers and constructors inherently benefited or suffered from the introduction of the sprint weekend format.
- `sprint_grid_size` accounts for the total number of drivers participating in a given sprint. It ensured the models accurately accounted for variations in the field size.
- `sprint_winner` is a binary feature indicating whether a driver won the sprint race. It was created by assigning a 1 to a first place sprint finish and a 0 to all other positions, serving as the target variable for the classification model.

6 Model/ Results

Grand Prix Predictions

To determine how effectively sprint performance translates to Sunday success, a Random Forest Regression model was developed to predict a driver's final Grand Prix finishing position based on their sprint results and weekend metrics. The model's hyperparameters were optimized utilizing a 5-fold GridSearchCV targeting the negative mean absolute error (MAE), which resulted in an optimal setup of 200 estimators and a maximum tree depth of five. To further ensure model stability, a minimum split threshold of five samples was also established. The regression model achieved a training MAE of 1.89 and a test MAE of 2.328, alongside a test R-squared value of 0.682. These results show that the model is able to reliably predict a driver's final Sunday placement within ≈ 2 positions. This confirms that sprint race performance is a highly significant predictor for the main Grand Prix.



Note the absence of data points for the 19th and 20th positions in the test set is a reflection of the 80/20 data split and the frequent occurrences of DNFs. In many races within this

subset, mechanical failures or crashes resulted in fewer than 19 drivers completing the full race distance.

To further build upon this, a secondary Random Forest Classifier was constructed to predict the outright winner of the Saturday sprint race, which is a necessary step to evaluate the weekend's impact on the Grand Prix grid formation. Given the class imbalance, with only 18 winners identified across 360 driver-race entries, the training and testing sets were stratified and a balanced class weight was applied. The final model demonstrated a high overall test accuracy of 93.1%. However, upon taking a closer look, the classification metrics reveal that this high accuracy was primarily driven by the model's ability to correctly identify non-winners, yielding a recall of 0.97 for that class. For the "Winner" class, the precision was 0.33 and recall was 0.25. These figures result in a low F1-score of 0.29 for the "Winner" class, highlighting that while the model is great at predicting which drivers will not win a sprint, it struggles in isolating the specific victor.

Train accuracy: 0.986				
Test accuracy: 0.931				
	precision	recall	f1-score	support
Non-winner	0.96	0.97	0.96	68
Winner	0.33	0.25	0.29	4
accuracy			0.93	72
macro avg	0.64	0.61	0.62	72
weighted avg	0.92	0.93	0.93	72

Random Forest sprint winner classifier, Confusion Matrix on test set

True label	Non-winner	Winner
	66	2
	Non-winner	Winner
Non-winner	3	1
Winner		
	Non-winner	Winner
	Predicted label	

The model's overall effectiveness can be best reflected by the macro average F1-score of 0.62. Compared to the weight average of 0.93, which is heavily skewed by the high number of "Non-winner" samples (68), the macro average treats both classes equally. This 0.62 macro average emphasizes the challenge of minority class prediction in imbalanced F1 data, where the high volatility at the front of the field makes isolating a single winner a considerable predictive

hurdle.

This difficulty in isolating the winner underscores the volatility at the front of the field, which directly affects the 2021-2024 Grand Prix grid configuration. When compared to normal qualifying weekends, the sprint format reshuffles the grid based on Saturday's outcome, frequently benefitting drivers with high raw pace who may have struggled in single lap qualifying. As identified in the exploratory analysis, this format specifically allowed drivers like Kimi Raikkonen and Nyck de Vries to secure significantly improved Sunday grid positions compared to their performance in standard qualifying sessions.

DNF Predictions

As the reference analyses did not explore DNF modeling in depth, the model algorithm, parameters, and feature selection were decided by our own exploration. We decided to use a XGBoost classification model to predict Grand Prix completion. The qualifying times were excluded, as their use would have restricted the study to post-1994 data. The model pipeline was engineered to process a heterogeneous feature set: categorical variables (Driver ID, Constructor ID, and Circuit) were handled alongside numeric inputs (Grid Position, Grid Size, Round Number, Season, and cumulative Team DNFs for the current season).

In choosing a model, the Random Forest, AdaBoost, and XGBoost algorithms were evaluated. While all ensemble methods demonstrated comparable results, XGBoost had a marginally higher accuracy. Hyperparameter tuning focused on tree depth, with values from one to five being tried. We settled on a maximum depth of three to optimize the bias-variance tradeoff. Deeper trees had significant overfitting while shallower configurations resulted in worse predictive accuracy.

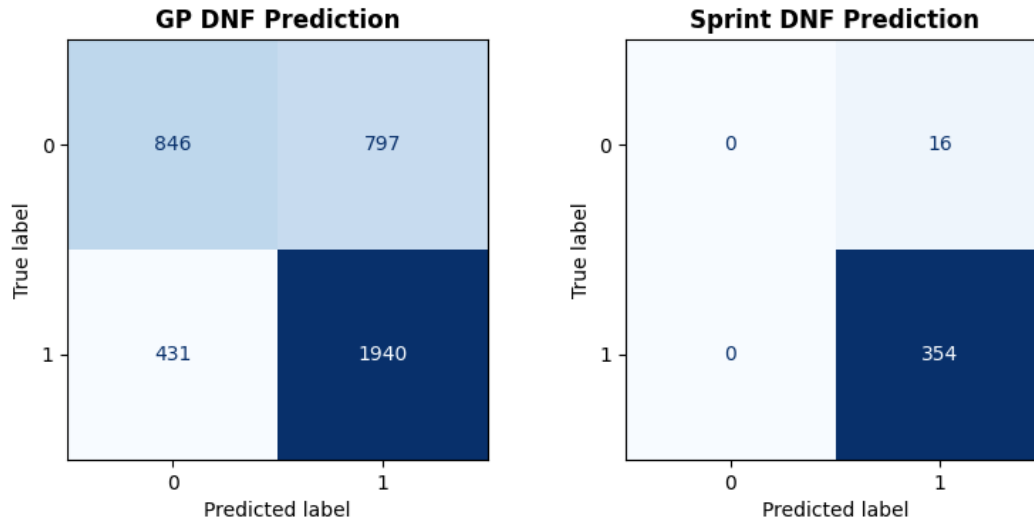
The loss metric used was negative log-loss. The final model had a training and test percent accuracy of 71.9% and 69.4%, respectively. The loss of the final model is 11.026, which is very good, considering the percent accuracy of the model. This low loss indicates that while the model does make mistakes it is very confident in the predictions that are correct and very tentative on the predictions that are wrong.

The model's ability to apply to sprint results was evaluated. There are not enough total DNFs in sprint races to train a model on it, but the data can be used as a comparison to see if some of the same factors that predict a DNF in a GP are the same as a sprint. The model had a very high success rate of 95.6% and a very low negative log-loss of 1.55.

As the figure below shows, the model only predicted finishes. This results in a high accuracy, but does mean that all true DNFs were mis-predicted. This is a safe guess for it to make, given the low DNF-rate of modern F1 races. This was not an entirely unexpected result based on two things: the smaller dataset size and that the sprint data is all fairly recent. The model has probably learned that recent races have low DNF-rates and adjusts accordingly.

Overall, we were able to determine that DNFs are not entirely uniformly random. Historically, DNFs were higher, mostly due to mechanical failures. The overall decline of DNFs have made modern incidents harder to predict. While the underlying distributions for Sprint and modern Grand Prix DNF rates seem comparable, the current 5% DNF rate creates a sample size challenge, making it difficult to confidently differentiate between the two formats.

XGBoost DNF Prediction Confusion Matrix



Sprint Predictions

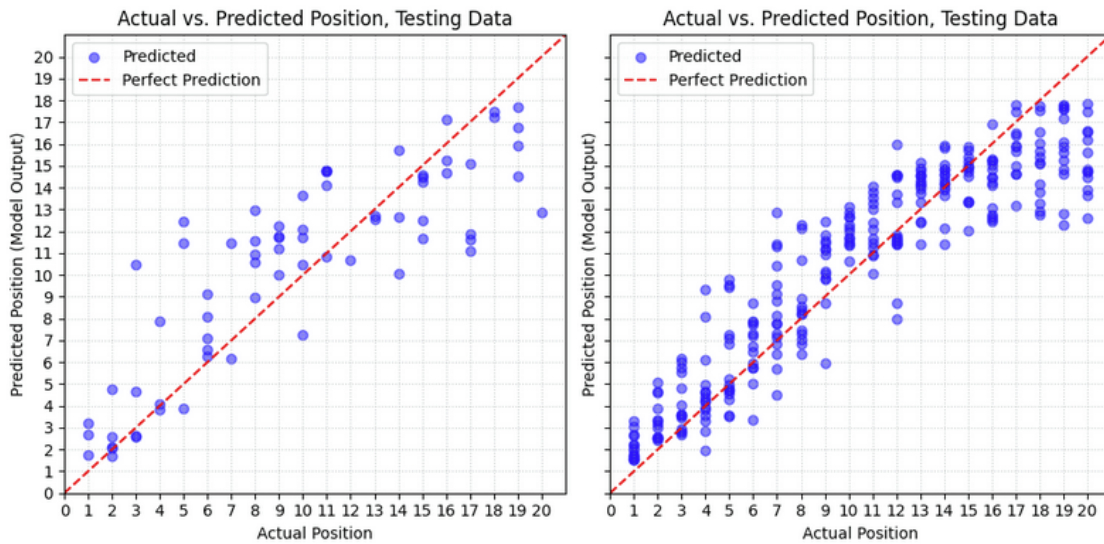
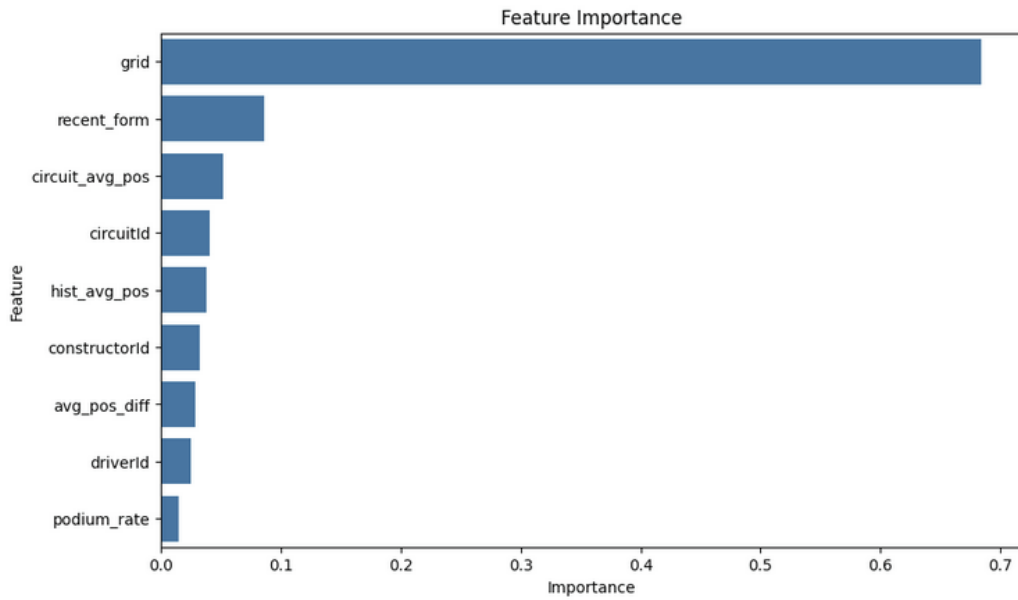
To predict the results of the sprint, models for regression prediction were tested. Regression was chosen over classification since the position of the drivers are not not unrelated. For example, the model predicting a position of 2nd for a driver that finished 5th is more accurate than the model predicting 20th.

3 Different models were tested for the prediction task: Random Forest, XGBoost, and Adaboost. Furthermore, 3 separate model training methods were tested. 1) Training on a dataset with only sprint data, 2) Building the dataset with features engineered from Grand Prix and sprint data, but training and testing only on sprint data. 3) Training on the entire dataset and testing only on sprint data.

The Random Forest model with training method 2 provided the best results. The predictive performance was measured with 3 regression metrics: mean absolute error, root mean squared error, and the coefficient of determination.

The model achieved an MAE of 2.28, which would be read as the model placing drivers within two positions of their actual finish. However, an RMSE of 2.97 is higher than the mean absolute error, and could be explained by outliers such as DNFs in the race. The R^2 value of .71 shows that a model covers 71% of the variance in the dataset.

To understand the logic behind the models predictions, the importance of the features was calculated, and is shown below. Unsurprisingly, the most impactful feature was the starting grid position for the drivers. With sprints being shorter than the Grand Prix, it logically follows that the start location would be the strongest factor to determine the prediction. Recent form was also a contributing factor. This feature captures the driver's current confidence and form, and also takes into account the constructors' technological abilities during the short period before the race.



The model's predictions vs the actual positions for the test set and training set are shown above. Something interesting to note is that the model did not pick any placements of 1 or 20. This could be explained by how Random Forest works. Since the algorithm takes the average of multiple decision trees, a few placements in the center could prevent the model from picking the upper and lower bounds of the race.

Overall, the sprint result predictions are fairly accurate, getting within a range of ≈ 2.32 positions of the actual position.

7 Code

Bartosz: <https://github.com/bartoszpro/DS675-Mini-Project/>

Patrick: <https://github.com/pdemarin/DS675-Mini-Project-Notebook>

Kunal: <https://github.com/kt398/F1-Predictions>

8 Video

https://drive.google.com/file/d/1dT5VRFsG0u_ad71S5QxYPKUUXd4Ovx3f/view

9 Milestone 2

https://docs.google.com/document/d/1y8_y3pwDXonCHKeHTLcvQSMpCjOU/edit?tab=t.0

References

- [1] <https://www.kaggle.com/datasets/rohanrao/formula-1-world-championship-1950-2020/>
- [2] <https://www.kaggle.com/code/jalelgmiza1/f1-race-qualifying-performance>